

CLASS 10 · CBSE · NCERT MATHEMATICS · DPP SHEET

Chapter 11: Areas Related to Circles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the area of a sector of a circle of radius 9 cm with angle 40° . Take $\pi = \frac{22}{7}$.
2. Find the area of a quadrant of a circle whose circumference is 44 cm.
3. The minute hand of a clock is 7 cm long. Find the area it sweeps in 15 minutes.
4. Find the length of the arc of a sector of angle 120° in a circle of radius 21 cm.
5. Find the perimeter of a sector of radius 10.5 cm and angle 60° .
6. A chord of a circle of radius 20 cm subtends a right angle at the centre. Find the area of the minor segment. Take $\pi = 3.14$.
7. For the circle in the previous question, find the area of the major sector.
8. A chord of a circle of radius 14 cm subtends 60° at the centre. Find the area of the minor segment. Take $\sqrt{3} = 1.73$.
9. A car has two wipers that do not overlap. Each wiper has a blade 21 cm long sweeping through 120° . Find the total area cleaned in one sweep.
10. An arc of a circle of radius 14 cm subtends 45° at the centre. Find the area of the sector and of the corresponding major sector.

Answer Key

1. $\frac{198}{7} \approx 28.29 \text{ cm}^2$
2. $r = 7 \text{ cm}$; area = 38.5 cm^2
3. $\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5 \text{ cm}^2$
4. 44 cm
5. $11 + 21 = 32 \text{ cm}$
6. $314 - 200 = 114 \text{ cm}^2$
7. $1256 - 314 = 942 \text{ cm}^2$
8. $\frac{308}{3} - 49 \times 1.73 \approx 102.67 - 84.77 = 17.9 \text{ cm}^2$
9. $2 \times 462 = 924 \text{ cm}^2$
10. 77 cm^2 and $616 - 77 = 539 \text{ cm}^2$